

An Article a Day: Towards Personal Informatics for Healthy News Consumption

Marianne Aubin Le Quéré*
msa258@cornell.edu
Cornell Tech
New York, New York, USA

Lauren Tran*
lat89@cornell.edu
Cornell Tech
New York, New York, USA

Andrew Berger
adb277@cornell.edu
Cornell Tech
New York, New York, USA

Christopher Nguyen
chris.nguyen@smartnews.com
SmartNews
San Francisco, California, USA

Rohini Venkatesan
rohini.venkatesan@smartnews.com
SmartNews
San Francisco, California, USA

Yuanhang Xu
yx376@cornell.edu
Cornell Tech
New York, New York, USA

Anastasia Sorokina
as2764@cornell.edu
Cornell Tech
New York, New York, USA

Mor Naaman
mor.naaman@cornell.edu
Cornell Tech
New York, New York, USA

ABSTRACT

Technological logs can enable us to self-reflect on our habits through data. Avid news consumers, who read many articles and take an interest in news media, may have a pronounced interest in tracking their news habits. From a lens of personal informatics, we asked twelve participants about their goals when consuming news. We visualized participant news consumption data in a dashboard to elicit self-reflections and desired actions. We find that avid news consumers are motivated by a desire to feel informed without sacrificing their mental health, dual goals which a news consumption self-tracking tool may help to achieve. However, misalignment between perceived news habits and trace data may lead to negative ruminations for users if their logged data reveals they read less serious news, less often, or less in-depth than their preference. We discuss the implications of these findings for the future development of personal informatics news consumption tools.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**.

KEYWORDS

news, quantified self, self-tracking, personal informatics, elicitation study, wellbeing

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*Both authors contributed equally to this research.

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1 INTRODUCTION

Healthy consumption and behaviors have been fostered through personal informatics tools aimed at tracking, monitoring, and adjusting personal behavior over time. Tools to encourage mindful consumption and lifestyles are most prevalent in the health and wellness space, where a plethora of personal informatics tools have been developed to drive mindful lifestyle choices (see [17, 24, 45, 49]). These personal informatics tools are defined as systems that “*help people collect personally relevant information for the purpose of self-reflection and gaining self-knowledge*” [32]. As personal informatics tools have become popularized, these tools are increasingly deployed to track media consumption habits [31], including reader behavior on Goodreads and music listening habits on Spotify [19, 50].

News consumption poses a specific challenge for personal informatics tools, since there is evidence that both under- and over-consumption of news can be detrimental to individuals. While the “informed citizen” ideal would posit that the public has a duty to stay informed about current events [10, 44, 55], research on news readership behavior finds that most people are underinformed about politics, local government, and basic knowledge [7]. Additionally, a recent body of research has found that overconsuming news can lead to negative mental health outcomes and news avoidance [1, 18, 39, 47, 51, 54]. Currently, there appear to be few paths forward for individuals to foster a sustainable, healthy relationship with news reading that still fosters maximum informedness.

While a rich literature has explored the perceptions and impacts of personal informatics tools in the health and wellness domains, scholars have called for the study of personal informatics tools in other domains [15]. We seek to understand the potential lived impacts of such tools in the domain of news consumption.

In this work, we specifically focus on how avid news users may respond to the development of personal informatics news tracking technologies. These users are the most likely early adopters of such tools, since news reading is a part of their identities [23] and they

may be intrigued to monitor their news consumption habits over time. We explore the goals and habits of avid news users, and how these users might *reflect* and *act* in response to seeing and tracking their own news data. The studies we ran leverage real aggregated user data to encourage participants to self-reflect on their news habits. This work provides insights into the future of news tracking tools, through a focus on the following research questions:

- How do the perceived news consumption habits of avid news readers diverge from their actual behavior?
- How do avid news readers reflect on the gap between their perceived and actual news consumption habits?
- How might avid news readers respond and change their behavior after reflecting on their news consumption?

2 RELATED WORK

In this section, we first build on studies seeking to understand people's new consumption habits; and then review studies in personal informatics exploring self-tracking in the news consumption space.

2.1 Understanding Digital News Consumption

As our news consumption habits have become digitized, researchers have been able to map and understand news consumption habits at scale with increasing precision through trace data and other methods [5, 30, 35, 36]. For example, “The NewsTracker,” a tool that records an individual's news site visits, illustrates how technological advancements now allow us to track news visits more closely [26].

These analyses of news consumption behaviors have led some researchers to define the avid news reader as a person who consumes news frequently relative to the baseline. Multiple studies focus on avid news readers that read for longer, often about more political topics, sometimes termed “news gourmets,” “news seekers,” or even “news junkies” [23, 28, 34]. Avid news readers consider keeping up with news part of their identity [23], and thus such readers may particularly care about evaluating or monitoring their news habits.

Studies have assessed where self-reports of media and news consumption diverges from logged data, finding that people largely overreport serious news consumption. Studies that have examined how closely reported internet and media usage mirrors logged data have tended to find low to moderate correlations [11, 40, 43]. Across different news media types, people tend to overestimate their own news consumption by three times, and are particularly likely to overreport political news consumption [20, 27, 41]. We focus here on understanding how avid news consumers make sense of and evaluate their own news consumption when exposed to their logged consumption data via personal informatics techniques.

2.2 Personal Informatics and News Consumption

Personal informatics research, the study of self-improvement and self-tracking, particularly through technology-enabled means, is an active area of research in the human-computer interaction community. The successful implementation of self-tracking habits is often motivated by personal goal-based objectives [37, 49]. There is some evidence that self-tracking can motivate positive behavior change [21, 45], and reciprocally that self-tracking itself is an outcome of motivations to change behavior [49]. Li et al.'s foundational

work on a stage-based model of personal informatics outlined five stages of any personal informatics tools: preparation, collection, integration, reflection, and action [32]. *Reflection* is the stage of engagement with self-tracking where people view, absorb, and integrate personal data into one's world view [33], while *action* is the stage at which self-trackers take action or make changes based on their data [32]. However, some have cautioned that not meeting one's goals can also lead to negative ruminations, when self-trackers feel defeated when looking at their data [12, 13, 25]. While personal informatics work tends to focus on health and wellness, scholars highlight the need to examine self-tracking in other domains [15]. We explore these themes in relation to news consumption.

Studies on the need for mindful news consumption motivate the application of personal informatics tools to help people maintain a “healthy news diet.” While there is a broad understanding that people wish to be informed about current events, informedness can look different to everyone [29]. A majority of people do not feel sufficiently informed [57], and the average person is civically underinformed [10]. In contrast, emerging largely from work on news consumption during COVID-19, studies also demonstrate that following the news too closely can have damaging psychological effects and even lead to news avoidance [1, 18, 39, 47, 51, 54]. Our study evaluates the potential impact of news consumption tracking to balance objectives for healthy news consumption.

There is some evidence that news readers would seek out a personal informatics system for news consumption, and some hypothesize that self-tracking could help news readers meet their goals [59]. Academics have engaged with self-tracking tools in the broader media landscape, including systems to track music listening, words or books read, or video game use [2, 33, 38, 42]. Scholars have experimented with tools that allow users to self-reflect on other online activities, including visited websites and search terms [3, 52, 58]. Most pertinent to the current study, Lee and Suh ran a field experiment where participants declared their intended reading topics, and showed that introducing self-tracking systems encouraged greater goal topic alignment, providing initial evidence that personal informatics concepts could assist news consumption goals [31]. In this work, we extend these findings by qualitatively exploring potential outcomes from engagement with a personal news tracking tool, using real participant data.

3 METHOD

To gain a better understanding of how avid news consumers might engage and react to a news tracking tool, we ran a study involving 12 elicitation interviews with heavy users of the news aggregator app SmartNews. During the interviews, we asked participants about their news habits, showed them a prototype that displayed their own news consumption data, and interviewed them about their reactions. Past lived informatics studies have shown how using real user data can foster rich insights from participants [14, 56]. All procedures were approved by Cornell University's IRB.

Our research site, SmartNews, is a news aggregator app that shows users news articles from many different news media outlets through one interface. The company says they combine human curation with machine learning algorithms to learn user interests and recommends news stories to users [46, 48]. The app has been

downloaded over 50 million times across the world, with over 1 billion page views per month [53]. We had access to SmartNews participant data via a collaboration and with the informed consent of participants.

For participant recruitment, heavy users of SmartNews were first sent a screener to confirm their use of SmartNews and that they were over 18 years old. We define heavy users of SmartNews as people who opened 50+ articles on SmartNews in one month, and who self-described SmartNews as one of their primary news sources. Twelve participants passed the screener and were interviewed by the research team; basic demographics are illustrated in the Appendix in Table 1.

During the interview, participants were shown an online dashboard with an aggregate view of their logged news habits over the past month on SmartNews, shown in Figure 1. The dashboard was an illustrative tool to elicit participant reactions to seeing their news data in aggregate. The dashboard visualizes the number of articles the participant read, sorted by topics and sources. In addition, the dashboard visualizes of daily article reading time on SmartNews; the average time per article by day of week; and articles read by time of day. We designed the dashboard iteratively, through balancing user needs that occurred during four pilot interviews and available trace data. For example, pilot participants wanted to see more time-based data, which led to us visualizing their news consumption over time in multiple ways.

We employed a semi-structured interview procedure with an emphasis on capturing participant perspectives about their news habits before, during, and after seeing the aggregate dashboard. Participants first answered questions about their motivations for reading the news, how reading the news fits into their daily routine, and whether participants were satisfied with their news habits. While participants were interacting with the dashboard, we used a think-aloud procedure to document their response to seeing their data. We also asked participants if the data they saw met their expectations. After participants had interacted with the dashboard, we asked them to re-reflect on their news habits, explain whether the dashboard had altered their perspective around their own news habits, and whether they may change their future news habits.

Interview data was analyzed using iterative inductive thematic analysis inspired by grounded theory [6]. Two authors open-coded two interviews each, including one overlapping interview. The research team met to discuss emerging themes and group co-occurring codes together. This process continued iteratively until the research team was satisfied that interviews could be comprehensively coded.

4 FINDINGS

In the findings section, we report on participant goals for news consumption, participant reflections while interacting with the prototype, and participant desires for future action.

4.1 News Consumption Goals and Habits

We discuss the duelling goals of participants who sought to be informed, yet struggled to manage their news consumption.

4.1.1 Need to be informed. All participants said a primary motivator for reading the news was to be informed. They described being informed as generally encapsulating reading news on politics,

economics, and other current events, in contrast to reading for entertainment purposes, which included lifestyle and celebrity news. Reading the news was a form of civic engagement for participants, who expressed desires to make informed voting decisions and participate in political discussions. P10 summed his motivations up succinctly as *to stay informed, and to make better decisions in my voting habits*. Three participants also expressed a desire to be a reliable source of information for others. For example, P17 stated: *“it at least keeps me ahead of the curve in terms of having something to discuss with others.”*

4.1.2 Managing news consumption. Five participants raised concerns about how reading news too closely or too frequently impacted their mental health, particularly for hard news topics or negative news. Four participants explained that they had a habit of avoiding negative news because reading led to anxious feelings. This challenge was sometimes hard to align with the feeling of responsibility to read news and stay informed. P15 stated, *“There’s times when I deliberately don’t read the news, because I don’t want to hear anymore about what’s going on around me. I find a lot of [news] kind of depressing and stressful these days. But I still normally read it.”* Participants commented that the type of news and the time of day that they consumed the news could impact their stress levels. For example, P8 said that reading the news is *“probably worse right before bed, because I get that feeling of sitting up and worrying about things.”* P12 was the most concerned about how their news consumption habits impacted their mental health, and described reading about serious news in the morning and if the news was particularly bad, they actively avoided news of that topic for the rest of the day to allow them to relax at night. Almost half the participants consciously balanced their need to feel informed with anticipated negative effects of reading the news on their mental health. Nine participants, including some who did not raise specific concerns about how the news impacts their mental health, still emphasized that checking the news is something they constantly do; *“it’s just something you do all day long,”* according to P11.

4.1.3 Reading the news as leisure. Five participants also expressed that reading the news could be a form of entertainment. P8 expressed that reading lighter articles was a personal reward: *“there are times, if I’m tired of ‘real’ news, I will treat myself to a little fluff piece. Like—oh look—some cute baby animal somewhere!”* P12 felt his news consumption was purely for pleasure, stating *“I’m not in school, no one’s grading me or anything, so I’m just trying to enjoy myself really.”* Participants also expressed that the news could be a way to pass the time, as P8 reported they’ll scroll through news *“if I’m waiting for a meeting at work, or I’ll scroll when there’s a TV show on in the background.”*

4.2 Reflecting on News Habits

We highlight participant reactions when perceived news consumption aligns or does not align with their logged data.

4.2.1 Initial reactions to logged data. Five of the participants expressed awe or joy in looking at their news habits through a data-centered lens. For example, P15 was surprised that individual news consumption could even be tracked and aggregated: *“I never really thought about [my news habits] this way, it’s completely new.”* Most

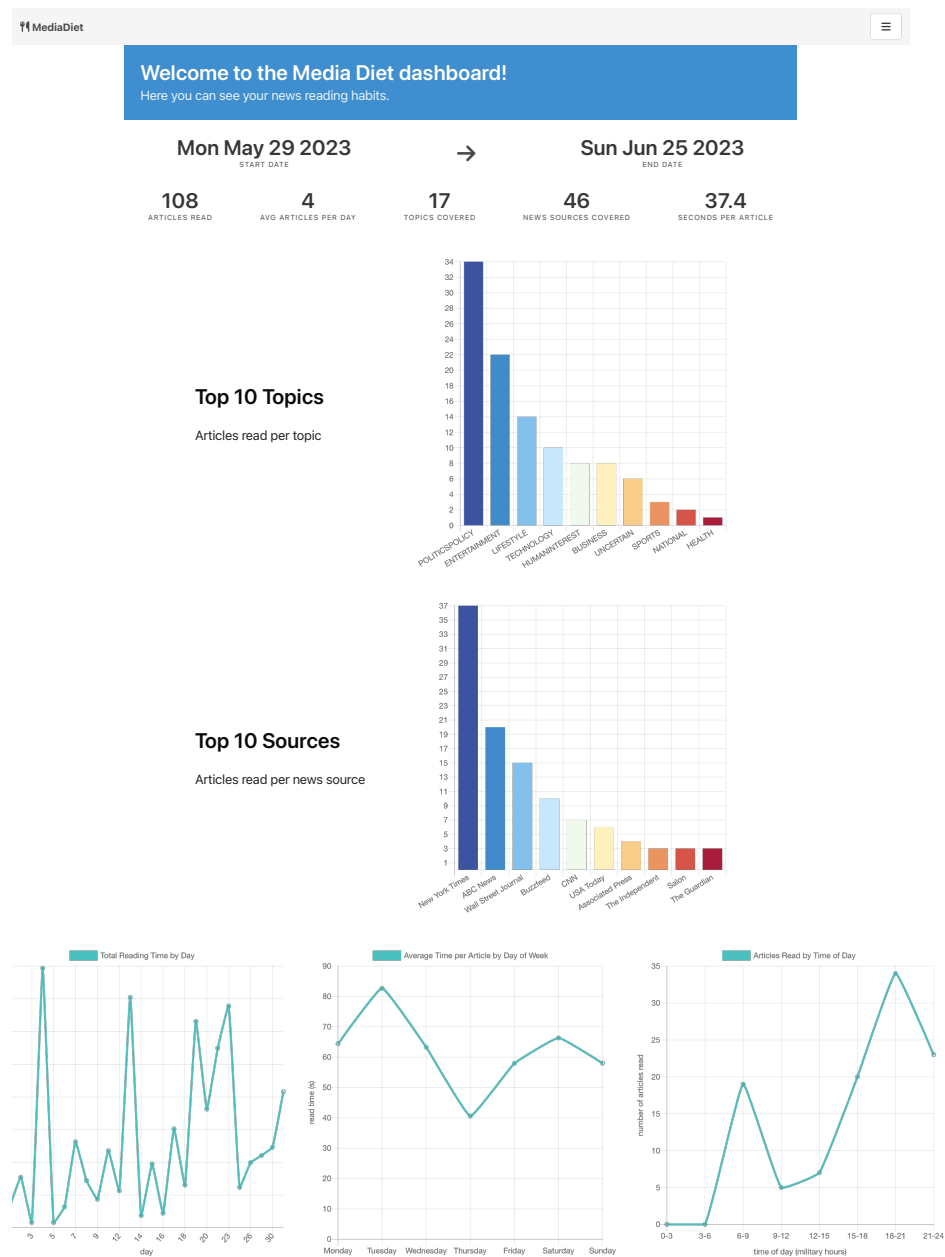


Figure 1: The dashboard presented to users with their aggregated news consumption data, populated with sample data. The top bar has summary information, such as the period the data was tracked during and how many articles the user had read.

participants expressed that perusing their own data was intellectually interesting. Reflecting this trend, P7 claimed that the dashboard was “*certainly accurate and helpful*,” while P4 thought data like this was “*interesting to see every now and then*.”

4.2.2 The gap between perceptions and logged data. The most prevalent theme that arose during the interview sessions was surprise by participants when shown the amount of entertainment news they

read in comparison to other topics. Out of 12 participants, 11 expressed surprise at their media diet dashboards, with eight of those participants reporting that the level of entertainment news was higher than expected. “*It does surprise me to see that I’m spending so much of my time on this useless stuff and not actual information*,” said P15, while P8 said he saw “*more lighthearted reading than I was expecting*.” Five participants with entertainment news as a top three topic expressed before seeing the dashboard that entertainment news was not one of their most read topics.

Overall metrics about the number of articles read or the time spent reading per article were also surprising to some participants. Three participants expressed that they thought they read more or for longer than the dashboard reflected, such as P16 who said *“huh... it’s about four a day... sometimes I feel like it’s more...”* Conversely, P15 thought that seeing all their data in an aggregated way added up to more than they realized.

4.2.3 Misalignment between perceptions and logged data. Participants who were surprised at the high level of entertainment news they read expressed negative sentiments. P5 found being confronted with the sources they read to be a sobering experience: *“when I see a bar graph like this, it really just kind of depresses me a little bit.”* When asked about emotions following their mention of surprise at entertainment news’ place on dashboard, P15 stated *“Oh my god, yes, I’m embarrassed... I shouldn’t be so interested in, like, crap that’s going on in other people’s lives as opposed to the actual news that occurring all around me.”* For these participants, viewing news reading data that did not match their self-perception led to negative ruminations.

Nine participants contextualized their news habits on Smart-News after viewing the dashboard. For example, four participants explained that their entertainment news consumption were articles they had shared with friends, or said they used other news platforms for serious news. P7, surprised at the low number of articles they read each day, explained that the data does not show the full picture: *“I thought I read more... but I get a couple of dozen of newsletters each day where I just glance through the headlines, so maybe that’s why.”*

Participants’ surprise was sometimes also followed by distrust of the way their data was collected and summarized on the dashboard. For example, P14 reported surprise at their top ten sources: *“I am surprised that I am following [source] as much as I am... I’m going to have to do some digging to validate those results and see how that’s coming about.”*

4.3 Future Actions

Finally, we report on how participants might change their behavior based on self-knowledge they gain from reflecting on their data.

4.3.1 Interest in ongoing monitoring. Most participants reported interest in news habit tracking in the future, despite low interest in changing their news consumption behavior. Ten participants were interested in viewing their news habits as an intellectual activity. Furthermore, many participants showed some interest in using news habit tracking to guide their behavior toward diversifying, improving, and monitoring their news consumption. When asked whether they would be interested in using a news tracking dashboard in the future, P14 said *“it would be an interesting dashboard to add to my own usage in monitoring what I’m looking at... you know, am I balancing [my news]?”* P13 similarly expressed the dashboard would *“almost serve as a double check on my self-analysis: Does this dashboard correlate with what I believe I’m doing, or is it radically different?”* P5 explained the dashboard could help them guide future news consumption decisions: *“it would be useful in pursuing more respected news content and more respectable sources.”*

4.3.2 From reflection to action. Five participants reported that seeing their media diet dashboard changed their self-perceptions about their news habits. Talking about how their thinking has changed, P6 said *“maybe it’s not a huge variety of news that I’m looking at—maybe that has changed—I thought that I liked a variety.”* For four participants this change in self-perception led to a desire to change news behavior by reading more diverse topics or more serious articles. For example, P13 stated he wished to *“find superior content, better-written content, less opinionated content, more long form content, less clickbait content.”* Three of the participants that wanted to change their actions found that viewing the dashboard made them feel shame or disappointment in their current news habits, and reversed from their stated pre-interview position that they felt no desire to change their news habits.

In contrast, seven participants said the exercise did not change their self-perceptions and would not guide future news behavior, even when the dashboard surprised them. When asked about whether the participant felt any different after viewing the dashboard, P12 said *“Not really. I’m a little surprised at the sources... but I’ve been in this routine for so long, I have no plans of changing.”* These participants expressed feeling set in their routines and did not alter their mindset, even when their perception of their news habits did not align with their logged data.

5 DISCUSSION

Motivated by recent work that has found the need for more mindful news consumption for an engaged citizenry [18, 39, 47, 51, 54], this study explored the possibility of using self-tracking tools to manage news consumption. In this section, we situate our findings relative to prior personal informatics work, contextualize our population of heavy news users, and provide a set of takeaways for future self-tracking systems for news consumption.

Our study shows how concepts from personal informatics literature map onto the news consumption domain. Participants exhibited the expected behaviors documented by Li et al. in different domains at the *Reflection* and *Action* stage [32] of the personal informatics tool we presented. Our participants expressed interest in monitoring their news habits out of intellectual curiosity, mirroring prior findings that people sometimes engage in self-tracking to create a personal archive [8, 17]. The downsides of personal tracking are also reflected in this study: prior work found that reflecting on unmet goals can lead to adverse psychological effects [9, 12, 16]. We find similar negative consequences for participants who engaged with more entertainment content than they wished, or who read fewer articles than expected.

Our interviews also exposed key dimensions where the news consumption domain may diverge from prior personal informatics literature. One such divergent dimension is that there is no established metric for the primary outcome that participants valued: *informedness*. Participants in our study conflated topic choice distribution with informedness, and thus valued topics like politics more highly than entertainment. Topic choice is not a standin for informedness, since there is evidence people still gain political knowledge through soft news [4]. While some end goals in the wellness space may be harder to track (e.g. reduced anxiety), most tools present established metrics such as calories exerted or heart

rate. Communication scholars highlight that there are a plethora of ways to perceive and measure informedness [29], and future work should explore which metrics people could reflect on to measure their informedness.

An additional difference in the news consumption domain is the need to balance over- and under-consumption of challenging news. While participants worried about informedness and associated hard news with certain news topics, some also reported having to manage when they read negative news to reduce stress. In contrast, a majority of personal informatics apps focus only on increasing a metric or hitting a certain target metrics, without need to counterbalance. Future work could seek to better understand how to develop goals from trace news data, and how to create personal news informatics tools that balance news consumption.

The findings demonstrate how avid news users, a “specialized audience” [22]), specifically may engage with self-tracking for news consumption. While participants were interested to examine their data, these avid news readers mostly did not want to change their news habits following our interview, perhaps since they already had healthy news habits [37]. Notably, these avid news readers perceived that they did not read enough hard news or spend enough time reading, suggesting that a more regular news consumer may be even less pleased when faced with their own news data.

Finally, we provide takeaways, based on our findings, to guide future personal news consumption tracking systems:

- (1) **Consider divergent, specific types of goals for news reading:** Though a majority of heavy news users have a general goal to be informed, success in this respect can vary by person. For example, people may wish to learn news faster, keep up with specific topics, or diversify their news reading.
- (2) **Be clear about how sub-metrics contribute to a larger goal:** There is no standard metric to measure informedness. When relying on easily calculable sub-metrics, such as the number of articles read, consider how these contribute to a larger goal.
- (3) **Make article logging explainable:** When aggregating data, provide clear instrumentation that shows how an article or behavior was aggregated a certain way.
- (4) **Leave room for error:** Most people do not read as much, as deeply, or as “seriously” as they wish. Consider how to present these gaps between ideal and reality in a productive, not discouraging way.
- (5) **Beware of the ceiling:** Keep in mind reading more news may not be universally desirable, especially if news are distressing or people read news late at night. Ensure news goals are not only centered around quantity.

Our study has several limitations. First, focusing on the data of users from one app means the dashboard is limited (“uni-faceted” [32]) in its representation of a participants’ total news consumption. We partially address this challenge by focusing on a news aggregation tool, instead of a single media outlet. Second, recruiting users who are heavy consumers of news resulted in a sample that skewed white and liberal, and limits generalizability. Future work may look at other types of news users, for example occasional news readers or news avoiders. Finally, future work may take a longitudinal lens,

and track whether usage of such a tool could impact news consumer behaviors over time.

6 CONCLUSION

In this work, we applied personal informatics literature to the domain of news consumption. Participants, who were avid news readers, were surprised by the mundane focus of the news they consumed, which led some to negative ruminations. While we expected this specialized population may have an acute interest in tracking their data, and they expressed an intellectual curiosity in ongoing monitoring, in most cases participants did not anticipate changing behavior. Scholars can build on this work to better define and consider how to measure being conscientiously informed.

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A PARTICIPANT SUMMARY

We include a table with participant descriptives in Table 1.

B PRE-SCREENER SURVEY

Thank you for your interest in this research! We are a research team at Cornell University who are working together with SmartNews to learn more about people's news habits. This is an interest form for users of SmartNews who are interested in joining the study.

- (1) What is your name?
- (2) How old are you?
- (3) What is your email address?
- (4) How would you characterize your use of SmartNews?
 - (a) SmartNews is my primary news source
 - (b) I use SmartNews as much as other news apps/platforms
 - (c) I occasionally use SmartNews
 - (d) I rarely/never use SmartNews
- (5) Are you interested in participating in this study? The study will be a 45 minute interview where we will ask you about

Table 1: Participant summary. Participants 1-4 were pilot interviews aimed to refine our interview protocol. All participants were located in the U.S.

Participant ID	Age	Gender	Hispanic/Latino	Race	Political Affiliation
P5	37	M	No	Black or African American	Center
P6	46	F	No	White	Center
P7	72	F	No	White	Liberal
P8	36	F	No	White	Very Liberal
P10	51	F	No	White	Liberal
P11	64	M	No	White	Liberal
P12	47	M	No	White	Liberal
P13	60	NB	No	Other	Center
P14	58	M	No	White	Conservative
P15	62	F	No	White	Liberal
P16	55	M	Yes	White	Center
P17	46	M	Yes	White	Liberal

your news habits and show you a prototype of an information tracking tool based on your SmartNews usage. You will be compensated \$20 in the form of a gift card for your time.

- (a) Yes, I am interested in participating.
- (b) No, I am not interested in participating
- (6) In order to participate in our study, we will need to access your data about how you've used SmartNews in the past 3 months. This may include any articles you've opened or read. All data shared outside of the direct research team will be anonymized and your identity will be protected. Do you give the research team at [Anonymous University] consent to access your SmartNews data for the last 3 months?
- (a) Yes
- (b) No

C PARTICIPANT SEMI-STRUCTURED INTERVIEW GUIDE

- (1) Why do you read the news?
- (2) How would you describe your news habits? (How do you usually read the news?)
- (3) How does reading the news fit into your daily routine?
- (4) Do you think there are better or worse times for you to read the news?
- (5) Do you have different types of news reading sessions?
- (6) Is there anything you would like to change about your news habits?
- (7) How do current events impact your news reading habits? Can you explain using a recent example?
- (8) Is there anything unexpected about your news reading habits?
- (9) Do you feel pride or shame about your news habits?
- (10) Do you read news from a variety of sources? A variety of perspectives?
- (11) How does SmartNews fit into your overall news habits?
- (12) Do you feel like this prototype accurately represents your use of SmartNews?
- (13) What do you find most surprising or interesting on the dashboard?

- (14) Is there anything else you would like to see on the dashboard?
- (15) On a scale of 1 - 5, with 1 being very dissatisfied and 5 being very satisfied, how would you rate your level of satisfaction with your news habits after having used the prototype?
- (16) How did seeing this prototype make you feel about your news habits?
- (17) Could you imagine using a dashboard like this? How would you use it?
- (18) How often would you check the dashboard and how would it fit into your daily life?
- (19) Do you think there's anything about your news habits that you'd want to change after seeing this dashboard?

D NEWS HABITS SURVEY

D.1 Demographic Questions

- (1) What is your age?
- (2) What is your gender?
- (3) Are you of Hispanic, Latino, or of Spanish origin?
 - (a) Yes
 - (b) No
 - (c) Prefer not to say
- (4) What is your race?
 - (a) White
 - (b) Black or African American
 - (c) American Indian or Alaska Native
 - (d) Asian
 - (e) Other
 - (f) Prefer not to say
- (5) Which best describes your political affiliation?
 - (a) Very Liberal
 - (b) Liberal
 - (c) Center
 - (d) Conservative
 - (e) Very Conservative

D.2 SmartNews Usage

- (1) Currently do you read news from many different news sources?
 - (a) Not at all
 - (b) Mostly not
 - (c) Somewhat
 - (d) A fair amount
 - (e) Very much
- (2) Currently do you read news about many different topics?
 - (a) Not at all
 - (b) Mostly not
 - (c) Somewhat
 - (d) A fair amount
 - (e) Very much
- (3) Currently do you read news from many different political perspectives?
 - (a) Not at all
 - (b) Mostly not
 - (c) Somewhat
 - (d) A fair amount
 - (e) Very much

- (4) How many news articles do you think you view daily? For the purpose of this survey, we define viewing as clicking on an article and reading at least a few lines.
- (a) 1-2
 - (b) 3-4
 - (c) 5-6
 - (d) 7-8
 - (e) 9+
- (5) What news topics are you generally interested in?
- (a) Science and Technology
 - (b) Politics
 - (c) World
 - (d) Entertainment
 - (e) Current Events
 - (f) Health
 - (g) Sports
 - (h) Education
 - (i) Environment
 - (j) Other
- (6) How much time do you spend on a piece of news (article) on average?
- (a) Less than a minute
 - (b) 1-2 minutes
 - (c) 2-5 minutes
 - (d) 5-10 minutes
 - (e) 10-20 minutes
 - (f) More than 20 minutes
- (7) How informed do you feel about current events in general?
- (a) Uninformed
 - (b) Somewhat uninformed
 - (c) Neither informed nor uninformed
 - (d) Somewhat informed
 - (e) Very informed
- (8) If you could change how informed you were about current events, would you choose to be...
- (a) Much less informed
 - (b) Less informed
 - (c) No change
 - (d) More informed
 - (e) Much more informed
- (9) I make dedicated time in my daily life to read the news to make sure I'm informed about current events
- (a) Disagree
 - (b) Somewhat disagree
 - (c) Neutral
 - (d) Somewhat agree
 - (e) Agree
- (10) News finds me: If something going on in the world is important enough, I'll hear about it somehow
- (a) Disagree
 - (b) Somewhat disagree
 - (c) Neutral
 - (d) Somewhat agree
 - (e) Agree
- (11) When I see a news headline, I always feel a need to click and read more
- (a) Disagree
 - (b) Somewhat disagree
 - (c) Neutral
 - (d) Somewhat agree
 - (e) Agree
- (12) When I read a news article, it's important to me to read it thoroughly
- (a) Disagree
 - (b) Somewhat disagree
 - (c) Neutral
 - (d) Somewhat agree
 - (e) Agree